

① Seven machines take 7 minutes to make 7 identical toys. At the same rate, how many minutes would it take for 100 machines to make 100 toys? (GATE - 2018) — 1M

- (a) 1 (b) 7
(c) 100 (d) 700

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2

Two coal loading machines each working 12 hours per day for 8 days handles 9000 tons of coal with an efficiency of 90%. While 3 other coal loading machines at an efficiency of 80% set to handle 12,000 tons of coal in 6 days. Find how many hours per day each should work.

- (a) 16 hr/day (b) 14 hr/day
(c) 15 hr/day (d) None

3

GATE-19: A contract is to be completed in 56 days if 104 identical robots work, each working at 8 hours a day. After 30 days, $\frac{2}{5}$ of the work is completed. How many additional robots should be deployed so that the work will be completed in the scheduled time, each robot now working 9 hours a day?

A) 160

B) 150

C) 24

D) 56

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- ✓ P, Q, R and S are working on a project. Q can finish the task in 25 days, working alone for 12 hours a day. R can finish the task in 50 days, working alone for 12 hours per day. Q worked 12 hours a day but took sick leave in the beginning for two days. R worked 18 hours a day on all days. What is the ratio of work done by Q and R after 7 days from the start of the project? (EC/ME – 2016) (2M)
- (a) 10: 11 (b) 11: 10
(c) 20: 21 (d) 21: 20

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GATE – 2020: (2M)

It was estimated that 52 men can complete a strip in a newly constructed highway connecting cities P and Q in 10 days. Due to an emergency, 12 men were sent to another project. How many number of days, more than the original estimate, will be required to complete the strip?

- (A) 3 days (B) 13 days
(C) 10 days (D) 5 days

40 men can complete a work in 48 days. 64 men started for the same work for x days. After x days, 32 men increased, so, the remaining work is completed in $16\frac{2}{3}$ days. Find x .

- (a) 5 (b) 8
(c) 10 (d) 6
(e) None of these

2 men or 5 women or 7 boys can do a piece of work in 469 days, then the number of days taken by 7 men, 5 women and 2 boys to finish the work is

- | | |
|--------|--------|
| A) 134 | B) 106 |
| C) 98 | D) 100 |

4 men and 6 women can complete a work in 8 days, while 3 men and 7 women can complete it in 10 days. In how many days will 10 women complete it?

- A) 35 B) 40
C) 45 D) 50

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①

An oil tank can be filled by pipe X in 5 hours and pipe Y in 4 hours, each pump working on its own. When the oil tank is full and the drainage hole is open, the oil is drained in 20 hours. If initially the tank was empty and someone started the two pumps together but left the drainage hole open, how many hours will it take for the tank to be filled? (Assume that the rate of drainage is independent of the Head) (GATE - 2019)

— 2M

- (a) 2.00 (b) 2.50
(c) 4.00 (d) 1.50

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A water tank is two-fifth full. Pipe A can fill a tank in 10 minutes and pipe B can empty it in 6 minutes. If both the pipes are open, how long will it take to empty or fill the tank completely?

- A) 6 min. to empty
- B) 6 min. to fill
- C) 9 min. to empty
- D) 9 min. to fill

✓ Through an inlet, a tank takes 8 hrs to get filled up. Due to a leak in the bottom, it takes 2 hrs more to get it filled completely. If the tank is full, how much time will the leak take to empty it?

- A) 16 hrs B) 20 hrs
C) 32 hrs D) 40 hrs

Three pipes A, B and C can fill a tank from empty to full in 30 minutes, 20 minutes, and 10 minutes respectively. When the tank is empty, all the three pipes are opened. A, B and C discharge chemical solutions P, Q and R respectively. What is the proportion of the solution R in the liquid in the tank after 3 minutes?

- A) $\frac{16}{11}$ B) $\frac{3}{6/11}$
C) $\frac{1}{6/11}$ D) $\frac{6}{11}$

A tap can fill a tank in 6 hours. After half the tank is filled, three more similar taps are opened. What is the total time taken to fill the tank completely?

- A) 4 hrs 45 min
- B) 3 hrs 45 min
- C) 5 hrs 45 min
- D) 5 hrs 45 min

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Three taps A, B and C can fill a tank in 12, 15 and 20 hours respectively. If A is open all the time and B and C are open for one hour each alternately, the tank will be full in?

- A) 4 hr B) 7 hr
C) 5 hr D) 6 hr

open/close (Join/left)

Two pipes P and Q can fill a tank in 6 hours and 9 hours respectively, while a third pipe R can empty the tank in 12 hours. Initially, P and R are open for 4 hours. Then P is closed and Q is opened. After 6 more hours R is closed. The total time taken to fill the tank (in hours) is ____ (GATE – 2019)

- A) 13.50 B) 14.50
C) 15.50 D) 16.50

* Two pipes can fill a tank in 20 and 24 minutes respectively and a waste pipe can empty 6 gallons per minute. All the three pipes working together can fill the tank in 15 minutes. The capacity of the tank is :

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- (a) 160 gallons
- (b) 240 gallons
- (c) 120 gallons
- (d) 180 gallons

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A booster pump can be used for filling as well as for emptying a tank. The capacity of the tank is 2400 m^3 . The emptying capacity of the tank is 10 m^3 per minute higher than its filling capacity and the pump needs 8 minutes lesser to empty the tank than it needs to fill it. What is the filling capacity of the pump ?

- (a) $50 \text{ m}^3/\text{min}$
- (b) $60 \text{ m}^3/\text{min}$
- (c) $72 \text{ m}^3/\text{min}$
- (d) None of these